

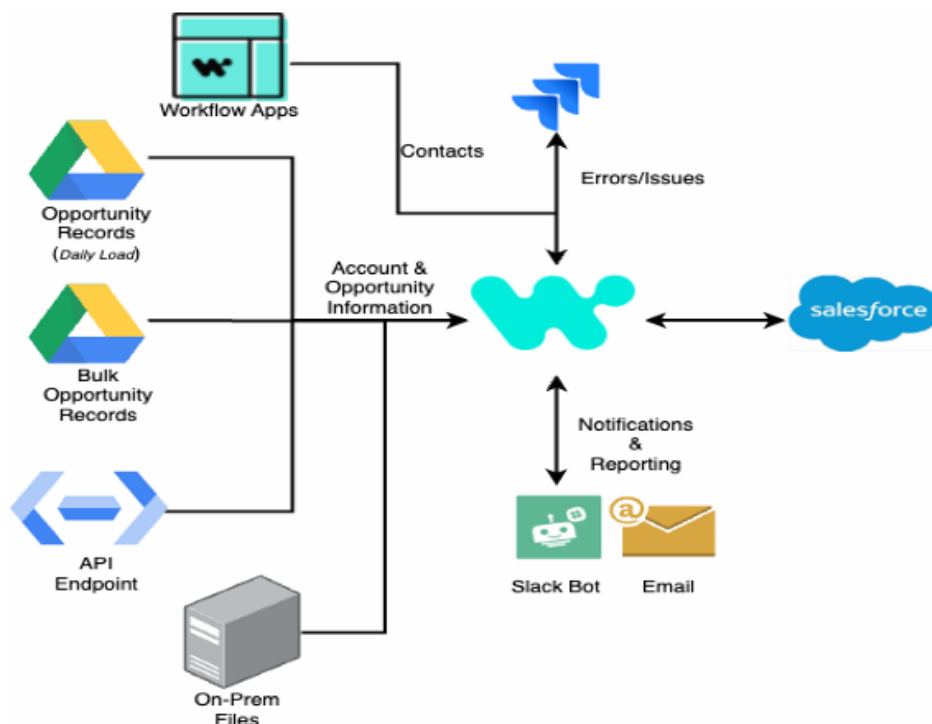
Customer Data Management Case Study

1. Introduction

ACME Corp is seeking to consolidate all customer and opportunity information into Salesforce CRM (SFDC) in a centralized manner. Currently, they acquire opportunities from various channels such as APIs, Files, and manual data entry into SFDC. However, the manual loading of all entries, including data from APIs and files, has resulted in data loss and errors. Consequently, the sales conversion rate has decreased due to delayed access to customer information for sales representatives. Additionally, ACME Corp aims to meticulously capture all errors and generate JIRA issues.

ACME Corp requires a solution that addresses the following objectives:

- Facilitates regular synchronization of customer data from multiple sources to SFDC.
- Provides real-time updates to sales representatives through Slack.
- Utilizes Workato's Workbot capabilities to extract CRM data for customized reporting within Slack.
- Generates scheduled reports via email, containing comprehensive contact data from Salesforce, filtered based on predefined criteria, along with relevant account information.



2. Exercise Prerequisites

- Create Slack Account

- Create a Slack Workspace from this [URL](https://slack.com/get-started#create):
 - (<https://slack.com/get-started#create>)
 - Enter your email (can be work or personal email)
 - Select a name for your workspace such as "<TeamName>_training_2024"
 - Note the name of your workspace as you will need it later on
- Refer to the [link](#) to establish a connection to Slack from Workato
 - <https://docs.workato.com/connectors/slack.html#how-to-connect-to-slack-on-workato>

- Create Salesforce Account

- Create a developer account in Salesforce [here](https://developer.salesforce.com/signup) (<https://developer.salesforce.com/signup>)
 - *Note- You can use your personal or official email address. Both works.*
- After account creation, you will receive an email in your inbox to access your account. Next, make sure you can log in to Salesforce with the newly created account using - <https://login.salesforce.com>
- Once the Salesforce account is ready, log in to Workato and create a Salesforce connection under your folder as given below:
 - Toggle the Green "Create" button and select Connection
 - For the Application, select **Salesforce**.
 - Name the connection however you want (e.g. John Smith Salesforce Account).
 - Select Location as a folder with **Your Name**.
 - Keep the Sandbox input field: No.
 - Click Connect
 - Enter the credentials of your Salesforce account and you are now ready to build your recipe.

- Create Google Drive Account

- Sign up for a new Gmail account [here](#). (<https://shorturl.at/jkJfv>)
- Using the same Gmail account make a connection to Google Drive
- Refer to the steps given [here](#) to create a Google Drive connection in Workato (<https://shorturl.at/cHoFF>)
- Create a folder named "Partner Delivery Training".

- Create JIRA Account

- Sign up for a new account here. Also, follow on-screen instructions, enter the desired site name, Take note of the site name, as this is needed for the hostname part for the Jira connection, etc
- Log in to JIRA with your newly created account as below: <https://<yourAccountName>.atlassian.net/jira/projects?selectedProjectType=software>
- After entering your email and site name, you are prompted with a few preferences (we can skip these), and then on the next splash screen, you get to choose the
 - 1.) template (Kanban), then the
 - 2.) Create a project screen, in which we can enter a relevant project name for this training session (e.g. Delivery Training), and click 'Create Project'
 - 3.) The Project board is now visible, w/ To-do, In Progress, Done
- Finally, in Workato, create a new JIRA connection using OAuth2.0 and enter your Jira Service Desk subdomain. Click "Connect" and you will be prompted to log into your Jira Service Desk instance using the account username and password in a separate window.
- Authorize Workato's request to access your Jira instance.
- If you need more details refer to the documentation [here](#) (<https://docs.workato.com/connectors/jira.html>)

3. Exercise steps for Workspace Management

- Create / Organize folders in your development env aligning to [GEARS best practices](#) for the Acme Corp use case. Refer to the table below for Asset codes.

Asset Type	Asset Code	Description
Recipe	REC	Core Workato recipe
Reusable Function	RF	Recipe without trigger meant for reuse across automation projects
Local Function	LF	Recipe without trigger meant for modular development
API Endpoint	API	Callable recipe exposed as an API
Connection	CON	Cloud App Connection (R, W, RW could also be used to determine Read/Write access level)
Connection (OPA)	OPA	On-prem App Connection (R, W, RW could also be used to determine Read/Write access level)
Lookup Table	LT	Lookup tables

Example –

Single workspace for all Teams/LoBs

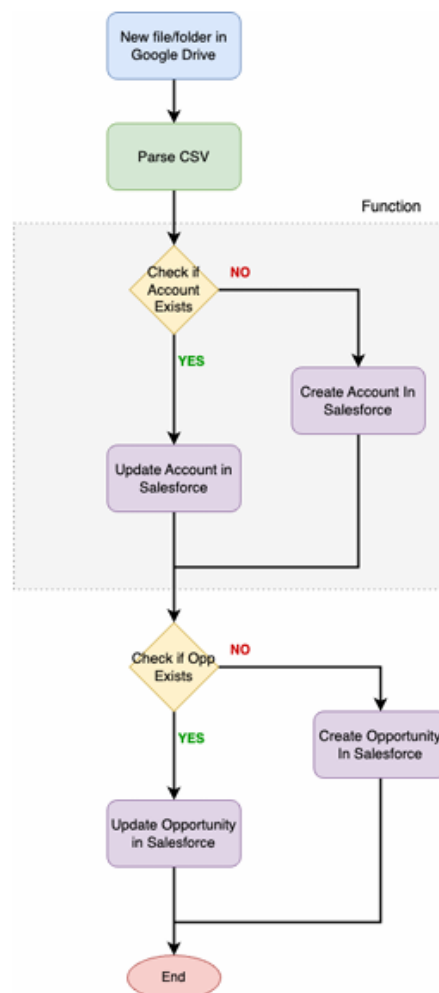
- Sales
 - Connections
 - [CDH] CON-RW | Salesforce
 - [CDH] Customer 360 Data Hub
 - API Endpoints
 - Recipes
 - [CDH] API | Get Customer 360
 - Bots
 - Slack
 - [CDH] BOT | Customer 360 workbot
 - Reusable Functions
 - [CDH] RF | Salesforce account lookup
 - Main
 - Recipes
 - [CDH] REC | Customer 360 Data Hub
 - Local Functions
 - [CDH] LF | Aggregate customer

4. Exercise steps for Developing Recipes

Description: This recipe will trigger whenever a new CSV file is uploaded to the specified Google Drive folder, parse the CSV data, search for or create an account based on the CSV data, search for an opportunity related to the account, and either create a new opportunity or update an existing one based on the CSV data.

- Based on the use case shared, create a recipe that triggers on a new file/folder from Gdrive. Next, it parses the CSV file from Google Drive and creates/updates Account and Opportunity to Salesforce.
- The flow Diagram for the recipe is shared below. Create a loop for multiple records.
- Also, ensure validation/creation of the account is invoked as a recipe function.

Sample CSV file can be downloaded here - <https://bit.ly/3ytr3dF>



5. Exercise steps for Error Handling

- Add Exception Handling to the above flow.
 - Add Monitor On Error Blocks
 - For each record in the CSV, if there is an error, retry at least once on error within a time interval of 5 seconds
 - Even after the retry If there is any failure, create an issue in JIRA
 - Notify the support team on Slack with the list of all Errors at once
 - For any other errors, create an issue in JIRA.
- Add comments to all steps, description and follow the best practices shared during the sessions till now like folder structure, naming conventions, etc.

6. Exercise steps for Task Optimization

Description: This recipe generates a report of all primary and secondary level Contacts and associated Account data in Salesforce CRM (SFDC) as an email attachment.

Problem:

- The [pre-built recipe](https://shorturl.at/4BIK3) (<https://shorturl.at/4BIK3>) shared has working functionality. However, the performance was never considered and the below issues are caused by the inefficient implementation of the recipe.
 - Higher total duration of the job
 - Higher total tasks used to achieve the requirement

With increasing contacts, the performance is even degraded and is not scalable in the current design.

Optimize the recipe:

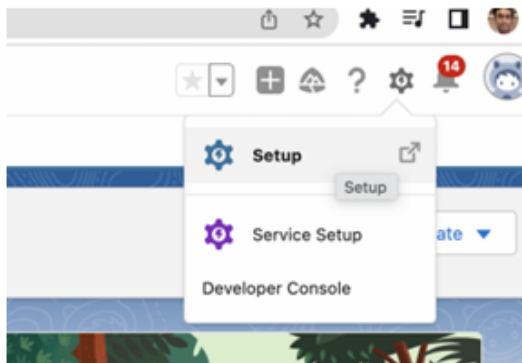
- Save the shared recipe into your project
- Configure the Salesforce connection to point to your instance.
- Configure the email address where the generated report is sent to.

- Test the recipe to observe the results in terms of job duration and total tasks consumed without any changes. Currently, it consumes about 97 tasks and the job duration is around 15 seconds.
- Optimize the recipe to run faster and consume an efficient number of tasks.

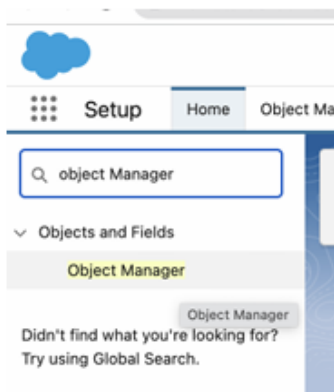
7. Exercise steps for ETL/ELT

Data Generation:

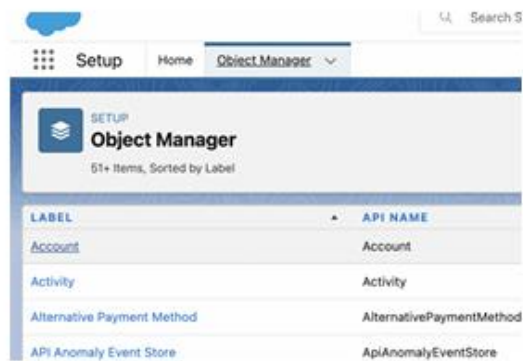
- Download the **GenerateAccountsData.csv** file from <https://bit.ly/3YwO8qF> and place it in the Google Drive “Partner Delivery Training” folder created earlier.
- Navigate to the below URL and use the recipe to copy to your project – <https://bit.ly/3FcsO2K>
- Login to Salesforce and go to Setup as shown below



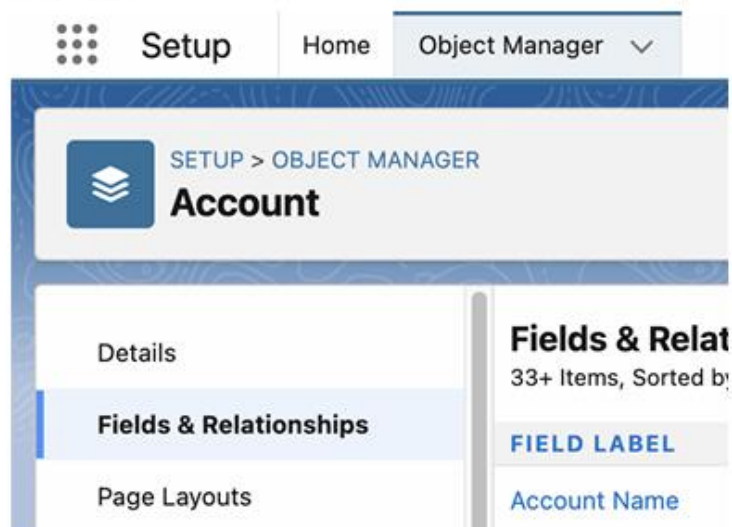
- In Quick Find available on the left, search for Object Manager:



- Within Object Manager, click on Account:



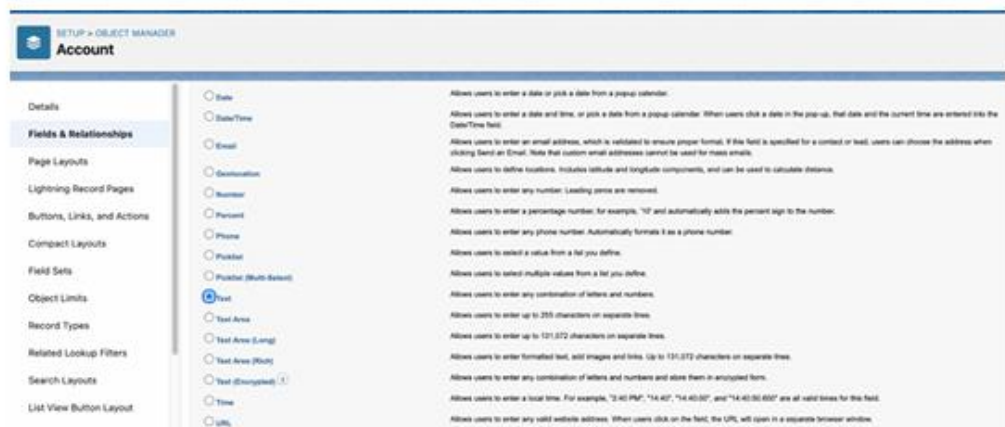
- Inside the Account Object, Navigate to Fields and Relationships as shown below:



- Then Click on the New Field option



- Select the Datatype as "Text" as shown below:



- Give the Field Name as “External ID”, length as 255 and select all the checkboxes as shown below:

The screenshot shows the 'Field Definition' form in Salesforce. The 'Field Label' is 'External ID'. The 'Length' is set to 255. The 'Field Name' is 'External_ID'. The 'Description' and 'Help Text' fields are empty. Under the 'Required' section, the checkbox 'Always require a value in this field in order to save a record' is checked. Under the 'Unique' section, the checkbox 'Do not allow duplicate values' is checked, and the radio button 'Treat "ABC" and "abc" as duplicate values (case insensitive)' is selected. Under the 'External ID' section, the checkbox 'Set this field as the unique record identifier from an external system' is checked. Under the 'Auto add to custom report type' section, the checkbox 'Add this field to existing custom report types that contain this entity' is checked. The 'Default Value' is set to 'Show Formula Editor'. At the bottom, there is a small text box with a formula: `Use formula syntax: Enclose text and picked values in double quotes: ("My Text"). Include numbers without quotes: (123). When percentages are decimals: (3.14), and express date calculations in the standard format: (Today() + 7). To reference a field from a Custom Metadata type record use: $CustomMetadata.Type__c.RecordName.Field__c.`

- Select all profiles and views in the next two pages and save the new field.

The screenshot shows two pages from the Salesforce configuration interface. The top page is 'Field-Level Security for Profile'. It has a table with columns 'Field-Level Security for Profile', 'Visible', and 'Read-Only'. The 'Visible' column has checkboxes for all profiles, and the 'Read-Only' column has checkboxes for all profiles. The bottom page is 'Page Layouts'. It has a table with columns 'Add Field' and 'Page Layout Name'. The 'Add Field' column has checkboxes for all page layouts, and the 'Page Layout Name' column lists the page layouts: Account (Marketing) Layout, Account (Sales) Layout, Account (Support) Layout, and Account Layout. At the bottom of the page, there are buttons for 'Previous', 'Save & New', 'Save', and 'Cancel'.

- Navigate back to Workato, open the recipe copied earlier, add your email address in Step number 5, update the connections to point to the Salesforce and Google Drive created earlier, and **start the recipe.**
- **The recipe will not work in test mode as you're using a long action (Upsert Accounts) in the recipe**
- Once you receive an email Stop the recipe and there should be a new file AccountsData.csv created in the same folder.
- This recipe should execute only once, after which this recipe should be stopped.

ETL/ELT Exercise

- Download the Opportunities CSV from here - <https://shorturl.at/rHyYO>
- Create a folder with your name in Workato File Storage and place the Opportunities CSV file.
- Load the Accounts and Opportunities CSV files, merge them using SQL Transformations, and load them to Salesforce. For more details refer to the flow shared below.
- Apply exception handling and failed records should be added to a JIRA issue and the support team to be notified via email.
- In the Bulk Upsert Opportunities action, use Name as the Primary Key as shown below.

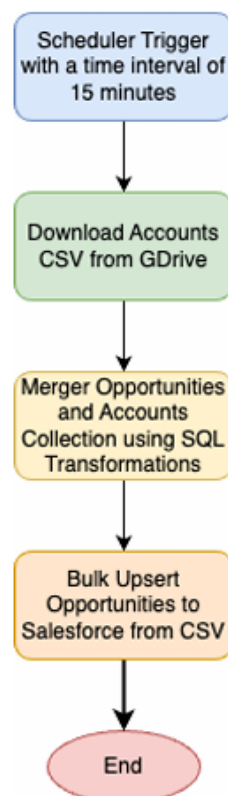
Primary key *

Enter primary key

ABC Name

Enter the API name of an indexed field (eg. Id for Record ID)

- The flow Diagram for the recipe is shared below.

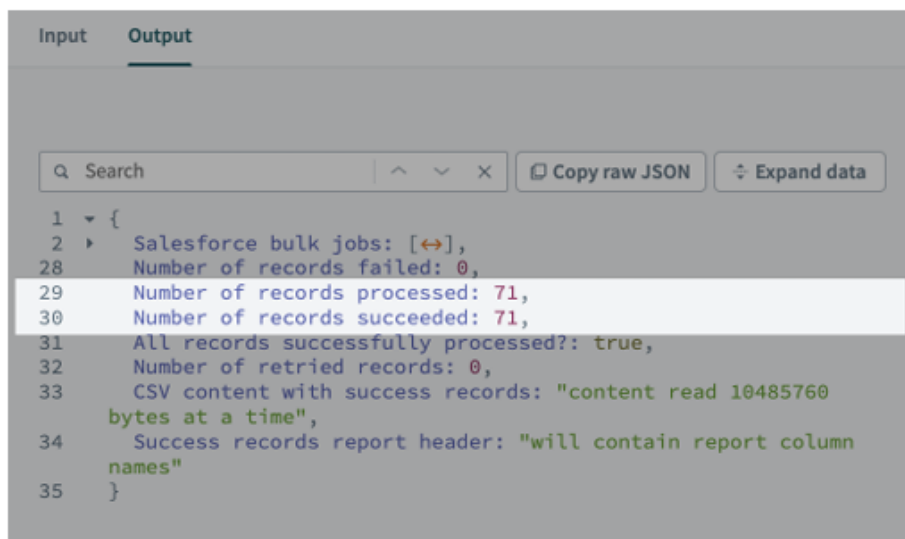


- **The recipe will not work in test mode as you're using long actions (Upsert Opportunities and SQL Transformations) in the recipe**

Sample SQL Query -

```
select opportunities.OpportunityName,
       opportunities.Stage,
       opportunities.AccountName,
       opportunities.CloseDate,
       accounts.AccountId
from opportunities
JOIN accounts on
       opportunities."AccountName" = accounts."AccountName";
```

- Also, validate if all records are processed in the Salesforce output. If there are failures, add an email step, and attach the CSV content with the failed record from the Salesforce step, and review the errors. Once fixed, rerun the job to ensure all records are processed.



ETL/ELT Additional Exercise

- ACME would like to use Workato to generate a report from their account's opportunity records.
- Generate a Google sheet report that calculates the total amount of money that needs to be paid by each account for all their opportunities.

B	C
acc_name	Total Amount
...	...

Note: Please use bulk action to add the records in Google Sheet

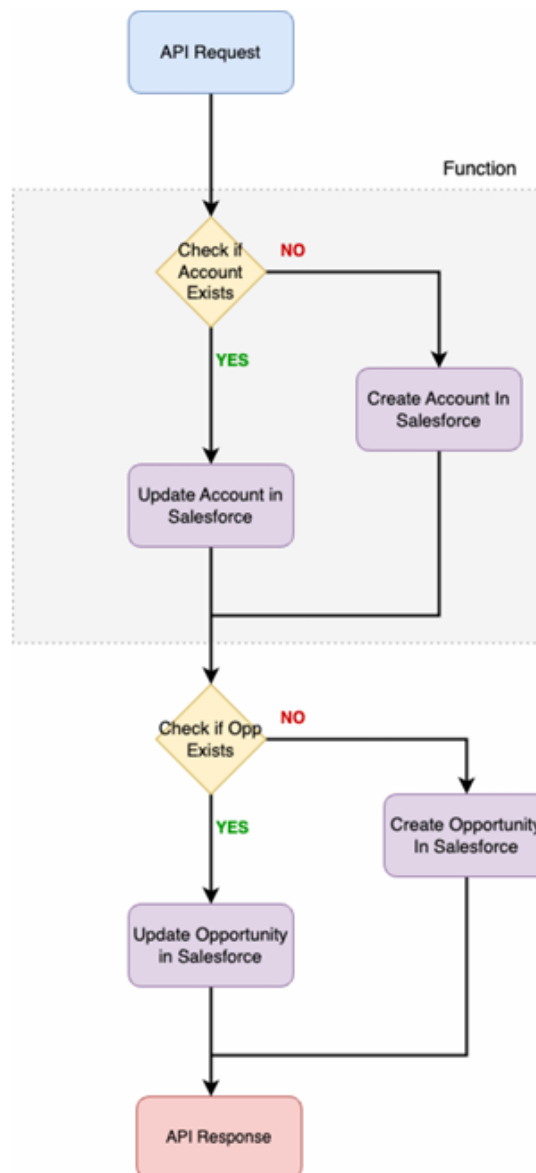
8. Exercise steps for API

- Clone the recipe created on [Day 1 \(Exercise 4\)](#) and modify it to expose it as an API as shown below.
- Create the below in our API Platform:
 - API collection
 - Policy which allows 5 requests/min and 100 requests/month,
 - A client with Auth Token Policy.
- Test the recipe from Postman/Workato API Platform and validate the below scenarios to verify if the data is added to Salesforce and the policy is working as expected.
 - Happy Path
 - Sending more than 5 requests per minute should not be allowed
 - Error Scenario

Useful Resources:

1. Postman can be downloaded from here - <https://bit.ly/3I7OXIA>
2. For Authentication using Auth Token, refer to the documentation here - <https://bit.ly/3ZDyH19>
3. Sample Postman Request here - <https://bit.ly/3ZD2mY5>

Flow:



9. Exercise steps for Test Automation

- Using your the recipe from [Day 1 \(Exercise 4\)](#) above, **create your first test case** for the recipe to test the flow for the Happy Path
 - Happy Path refers to the successful scenario where the opportunity data in the CSV are upserted to Salesforce.
- Navigate to the Test Cases page in your recipe and start creating a new test case.

Projects > Delivery Bootcamp > Sales/[CDH] Customer Data Hub/M... > [CDH] REC | Insert/Update Opportunity to Salesforce... : Test cases

[CDH] REC | Insert/Update Opportunity to Salesforce from CSV (Exercise 4 & 5 Updated)

Recipe Jobs Connections Versions **Test cases** Settings

Run test case Delete + New test case

Name	Modified on	Last run	Last run recipe version	Last run date	Case summary	Description
Happy Path	Oct 18, 2024 10:41 PM	Passed	6 Current	May 7, 2024 4:46 PM	1 mocked step 2 checks on 2 steps	

AssistIQ

- Use mock trigger data from a previous job that has the data that you are testing for.
 - If you have a previous job with suitable opportunity data to be upserted to Salesforce, use that job's trigger output to mock your trigger data.
- Mock steps & create checks in the recipe as needed.
 - Take note that **recipe functions** in your recipe have to be mocked with sample output.
- Verify the recipe steps work as expected by running the test case!

10. Exercise steps for Bots

- Create an enterprise Workbot with the name Salesforce Bot in your Slack account and enable Home Page
- Home Page should be designed as below

Salesforce Bot

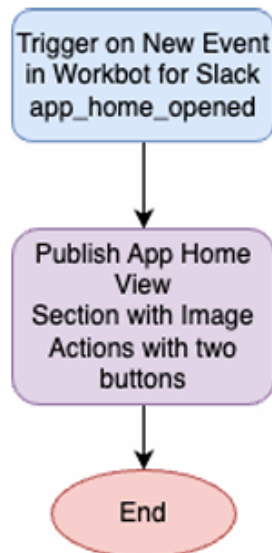
Home Messages About

Hi @partnersalesdemo, Salesforce Bot at your service. How can I help you today?

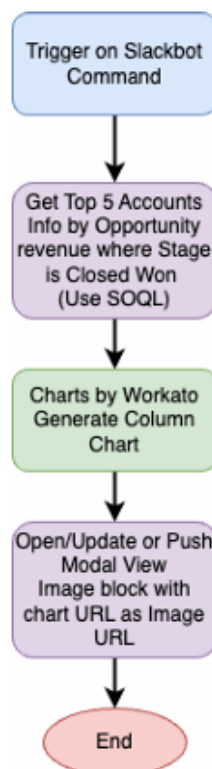
Click on the below options to obtain Salesforce related reports

Top 5 Accounts Top 5 States

salesforce



- Create a recipe to obtain the top 5 accounts based on Revenue and display the reports in a popup.
- Use the below flow in the recipe



- Hint: Use the below SOQL for the top 5 accounts -

```
SELECT SUM(Amount) Total, Account.Name FROM Opportunity WHERE StageName = 'Closed Won' GROUP BY Account.Name ORDER BY SUM(amount) DESC
```

11. Exercise steps for Workflow Apps

Description:

- Acme Corp aims to offer a solution for external and internal users without CRM access to input contact information.
- The entered details should undergo a review process before being added to the CRM system, encompassing leads, vendors, or suppliers for Acme Corp.
- Develop a submission form accessible both within and outside the organization, and implement automated contact creation following approval.

Steps

1. Create a new project with the following details:
 - **Name:** "CRM_Contacts_YourName" (*No Spaces or special characters*)
 - **Project Description:** Create contacts in CRM after approval
 - **How do you want to start?:** Start with a Workflow App
 - **App Name:** "CRM_Contacts_YourName" (*No Spaces or special characters*)
 - **The remaining Options as shown below:**

What do you want this workflow app to do?

- ☒  **Submit requests**
Collect information needed for further processing
- ☒  **Review requests**
Approve or reject any submitted requests

☐ Use an existing table to store app data

2. This will create a data table as well as two pages and a recipe for you.
3. Navigate to the data table tab and modify your data table as shown below.
 - Rename Name column as FirstName

- LastName, Email, and Company are 3 new short text fields
- Description column is deleted
- Reviewer Comments is renamed as Comments.

CRM_Contacts_<YourName> [View activity](#)

Last edited November 3, 2023 at 4:00 AM

3 hidden columns [+ Add filter](#)

Showing 0 records

	FirstName *	LastName	Email	Company	Comments
+					

- Now navigate back to your project and select the Assets tab.
Click on the page **Submit request in CRM_Contacts_<YourName>**

- Delete Description & Name fields.
- Drag FirstName, LastName, Email & Company fields to the page.
- FirstName, LastName, and Email fields are to be set as required.
- Modify the main page description as given below:
"# 🚀 Request creation of a new contact in CRM"
- Save the changes.
- The page should finally look like this:

🚀 Request creation of a new contact in CRM

FirstName *

LastName *

Email *

Company

Submit

5. Now navigate back to your project and select the Assets tab.
Select the recipe **"New request in CRM_Contacts_<YourName> will assign a task to users and wait until completed"**
6. Click on Edit Recipe and navigate to Step number 2. Modify the Task Name and Assignee as shown below. Select Optional Fields and select Page and choose the **Review request in CRM_Contacts_<YourName>** page in the dropdown. The workflow stage is to be set to In progress.

The screenshot shows a configuration window titled "Assign a task to users in CRM_Contacts_ and wait until completed". At the top, there are tabs for "Find", "Show optional fields", and "Group map data". Below the tabs, a note says "remain the same." The main configuration area includes:

- Task name ***: A text input field containing "Review FirstName Step 1 contact creation in CRM". Below the field is a description: "Describe what needs to be done. Task name will be displayed in email subject lines. Use datapills to personalize it."
- Assignee ***: A dropdown menu set to "Use email address or group ID" and a text input field containing "<your Email Address>". Below the field is a description: "Provide the email address or group ID to assign the task to."
- Page**: A dropdown menu showing "Review request in CRM_Contacts_<YourName>". Below the field is a description: "Choose from existing pages or [create a new page](#) for assignees to complete the task in."
- Number of days to complete task ***: A dropdown menu set to "Choose from list" and a text input field containing "1 day". Below the field is a description: "Tasks expire if they aren't completed within the specified timeframe."

7. Add an Upsert Contact in Salesforce Step before Step number 4. Use Email as the Primary Key.
 - Map FirstName, LastName, and Email from Step number 2. *If you do not see all the new fields added to the data table in Step 2 Output, click on the Refresh button in the top right corner of the recipe once to see all the new fields*

Upsert Contact in Salesforce

Find Show optional fields Group map data

Primary key *

✕ Email

Select field to search for record by. This field should uniquely identify records, e.g. characters ID. If record is found, it is updated. If record isn't found, it is created.

Last Name *

ABC  **FirstName Step 2**

Email

ABC  **Email Step 2**

Search for where Email has this value. If a record already exists, it will be updated. Or will be created

Account ID

ABC

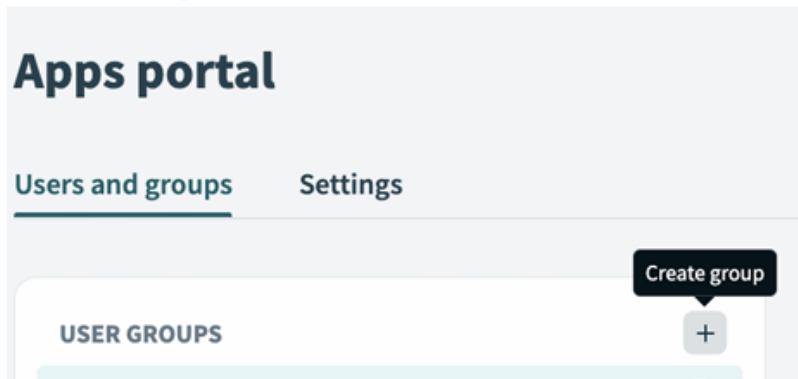
First Name

ABC  **FirstName Step 2**

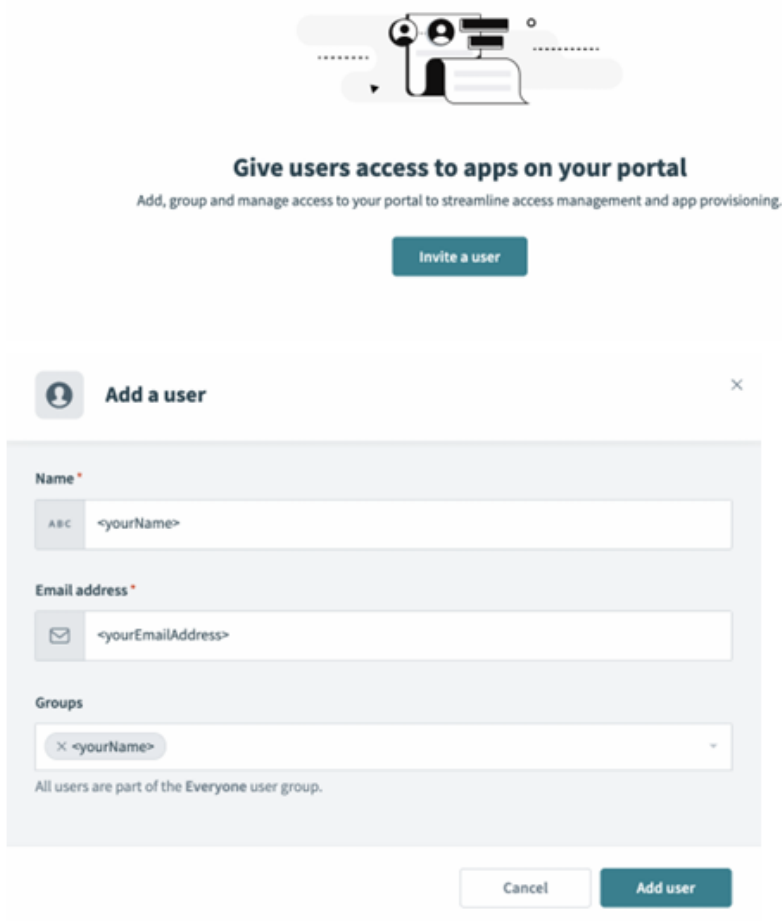
The final recipe should look like the one shared below. Save the changes and start the recipe.



8. Navigate to Platform-> Workato Apps Portal. Create a new Group with name as <yourName>



Once the group is created, select the group and click on Invite User to Invite yourself to the Apps Portal.



9. Navigate back to your project and select the **Assets** tab.
Click on the page **Review request in CRM_Contacts_<YourName>**
- Delete Name and Description fields
 - Drag FirstName, LastName, Email, Company, and Comments fields to

the page.

- All fields are to be set as read-only, except for Comments.
- Modify the main page description as given below:
"# 🛠️ Review contact creation request"
- Save the changes.
- The page should finally look like this:



🛠️ Review contact creation request

FirstName

LastName

Email

Company

Comments

10. Navigate back to your project and select the **Apps** tab. Click on Manage Access and select your group created with your name and role as Manager.

**CRM_Contacts_<yourName>** 
• Offline |  No user group added

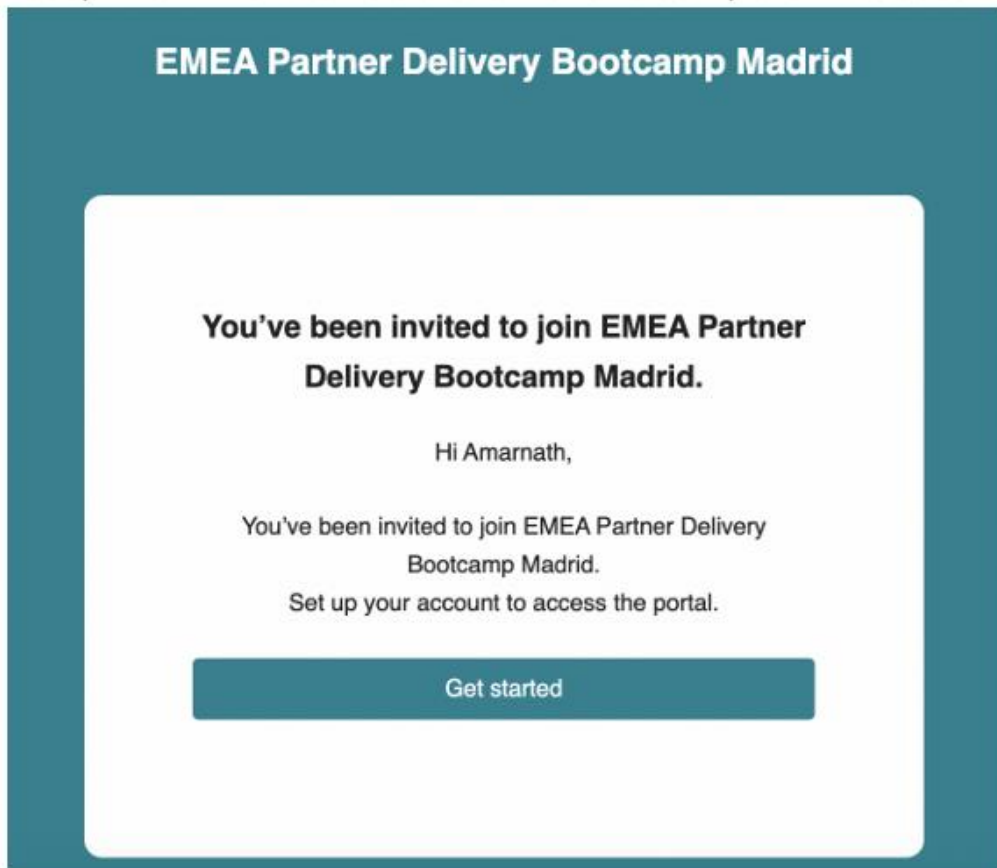
 **Manage access** ×
Give user groups permissions to access and use this app. To edit user groups, go to [apps portal](#) .

User group	Role
<yourName> (0 users) To be added	Manager  

11. Finally, click on Go Live.

Exercise Review

1. Accept the Invite received over email and complete the account setup



To access the portal, set up your account.

Name

Amarnath

Email address

amarnath.munkur@workato.com

Password

Password

+ At least 8 characters + Upper & lowercase letters + Number
+ Special character

Confirm password

Confirm password

+ Passwords match

Continue

2. You will be redirected to the portal after you select continue, If not log in to the portal using the link provided here - <https://initraining.workato.app/portal>. Use the credentials created in Step 1.

Hi Amarnath, what would you like to work on today?

Your apps



CRM_Contacts_Amarnath

3. Select your app and click on New Request. The submission form should be displayed.

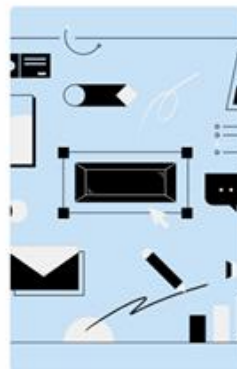


CRM_Contacts_Amarnath

All requests Assigned to me Created by me Due today

Create new request

CRM_Contacts_Amarnath > Create new request



Request creation of a new contact in CRM

Firstname *

Lastname *

Email *

Company

Submit

4. Make sure the recipe created as part of the exercise is started.
5. Submit a new contact request using the form. After submission, you should be able to see the below request.

CRM_Contacts_Amarnath

All requestsAssigned to meCreated by meDue today

Create new request

First Name	Current stage	Current task	Assigned to	Due on	Created by	Created at
John	In progress	Review John contact creation in CRM	Amarnath (me)	9 Nov at 3:40PM	Amarnath (me)	8 Nov at 3:40PM

6. You should also receive an email stating that the approval task has been assigned to you. Since you have created your user and assigned Manager role, you can approve your own submission.

SampleDemoApp <admin@workato.app>

3:53 PM (13 minutes ago)

to me

EMEA Partner Delivery Bootcamp Madrid

The task [Review Mike contact creation in CRM](#) due on February 17, 2024 has been assigned to you.

Go to task

7. Click ok Go to task to approve the contact creation.

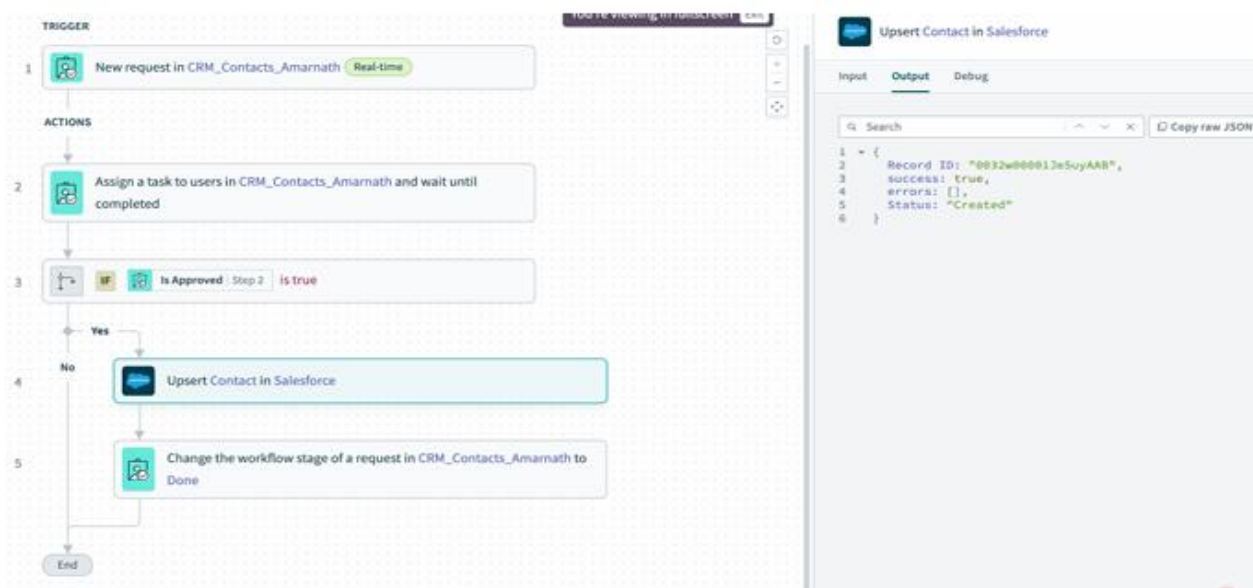
8. Once approved, you should see the task status being updated to Done. Also, review the recipe jobs to validate the contact creation.

CRM_Contacts_Amarnath

All requests
Assigned to me
Created by me
Due today

Create new request

FirstName	Current stage	Current task	Assigned to	Due on	Created by	Created at
John	Done	—	—	—	Amarnath (me)	8 Nov at 3:40PM



12. Exercise steps for OPA

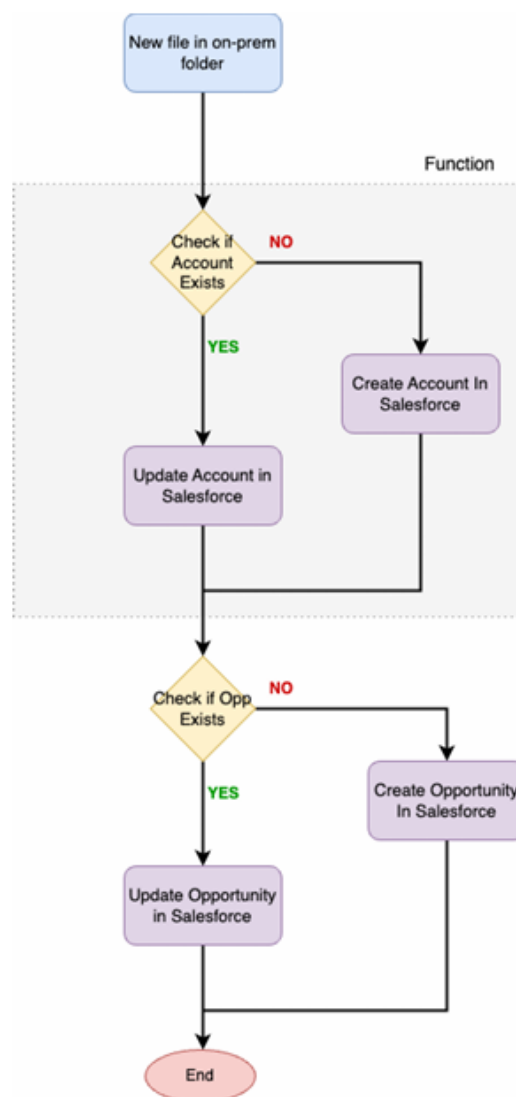
- Clone the recipe created on [Day 1 \(Exercise 4\)](#) and modify it to read the file from **On-Prem folder** instead of Google Drive.
- To setup OPA in your systems, refer to the below document <https://docs.workato.com/on-prem/agents/setup.html>

In Set up connections* choose **“Directly in Workato”**

Setup an On Prem File connection using the below instructions

<https://docs.workato.com/connectors/on-prem-files.html#how-to-connect-to-on-prem-files-on-workato>

- The flow is shared below.



13. Exercise steps for RecipeOps

Description: The leadership team at Acme Corp would like to receive notifications about the Total Active Recipes, Inactive Recipes, Connections, Allotted Tasks, and Available Tasks count at an account level in Slack on a daily basis.

 **workbot** APP 22:13
Account Details

Active Recipes: 4

Connections: 7

Available Tasks: 2999964

Inactive Recipes: 21

Alloted Tasks: 3000000

Also, the Project Manager would like to receive a daily CSV report of the recipe that uploads CSV to Salesforce (*recipe created on Day 1*). The report should contain the below information.

- Job ID
- Recipe ID
- Status
- Started At
- Completed At
- Time Taken per Job

JobReport.csv						
	A	B	C	D	E	F
1	Job ID	Recipe ID	Status	Started At	Completed At	Time Taken per Job
2	j-AHDFehdJ-AYwW9k	16300761	succeeded	2023-05-22T23:42:3	2023-05-22T23:42:4	3
3	j-AHDFdwFF-fDX3eY	16300761	succeeded	2023-05-22T23:41:3	2023-05-22T23:41:3	3
4	j-AHDFdnW6-AhemH	16300761	succeeded	2023-05-22T23:41:1	2023-05-22T23:41:2	3
5	j-AHDFdTX6-WHfLDC	16300761	succeeded	2023-05-22T23:40:5	2023-05-22T23:40:5	3
6	j-AHDFdHtW-kJaKBg	16300761	failed	2023-05-22T23:40:3	2023-05-22T23:40:4	5
7	i-AHBCkbf-rms8rhz	16300761	succeeded	2023-05-19T01:03:3	2023-05-19T01:03:3	3

Use the below flow in the recipe -



14. Exercise steps for HTTP Connector

Description: The internal development team at Acme Corp uses Todoist to manage and work on issues/errors etc. They even use todoist for regular updates on issues and day to day activities.

Problem:

- No Platform connector for Todoist
- The custom connector which is available, currently only supports API Key authentication.
- Acme Corp would like to connect to Todoist using the Oauth2 Authorization Code grant which is standard / secure.

Exercise Steps

- **Sign up for a free** Todoist account using the below link
<https://app.todoist.com/auth/signup>
- After completing sign up, **Log in to Todoist Developer** Console: Go to the Todoist Developer Console by navigating to <https://developer.todoist.com/appconsole.html>.
- **Create a New App:** If you haven't created any apps before, you'll need to register a new app. Click on the "Create new app" button under the Manage App page.
- **Fill in App Basic Info:**
 - App Name: Provide a name for your app.
 - App service URL : Enter your app's website URL (if applicable).
- **Create App :** Click on the "Create App" button which opens up the details page.
- **Fill in App Details:**
 - OAuth Redirect URL: This is the URL where users will be redirected after granting access to your app. The value for this should be <https://www.workato.com/oauth/callback>
 - Description: Describe the purpose of your app.
- **Save the app :** Click on the "Save settings" button to save your app.
- Take Note of Client ID and Client Secret: Once your app is created, Todoist will provide you with a Client ID and Client Secret. Keep these credentials secure as you'll need them to authenticate your app.
- Create an HTTP connection to Todoist using OAuth2 Authorization code grant using the below information:
 - Authentication type - [OAuth 2 \(authorization code grant\)](#)
 - OAuth2 authorization URL - <https://todoist.com/oauth/authorize>
 - OAuth2 token URL - https://todoist.com/oauth/access_token
 - Client ID & Client Secret - *created in the earlier steps*
 - Base URL - <https://api.todoist.com/rest/v2>
 - Advanced Settings -> Scopes -
[task:add,data:read_write,data:delete,data:read,project:delete](#)

- Create a new recipe and test the HTTP connection for two actions.
- Refer to Todoist REST API documentation [here](https://shorturl.at/leuQy) (<https://shorturl.at/leuQy>)
- First create a new project using the below JSON and the URL.

Relative URL - /projects

METHOD - POST

JSON -

```
{
  "name":
  "CompanyName_GroupNumber_DeliveryTraining"
}
```

- Then create a new task in the project created above using the below JSON and URL.

URL - /tasks

METHOD - POST

JSON -

```
{
  "content": "Partner Delivery Training HTTP
exercise",
  "due_string": "tomorrow at 12:00",
  "due_lang": "en",
  "priority": 4,
  "project_id": "<From Create Project step output>"
}
```

15. Exercise steps for Open API Connector

- Install the Open API Connector from the Community library
https://app.workato.com/custom_adapters/50046/details?community=true
- Create a new Open API Connection using the below YAML file and settings shared below
 - Server URL - <https://api.github.com>
 - Open API Document URL - <https://shorturl.at/npPcJ>

- Authentication method – Basic
- Username – *Your github userID*
- Password – *Personal Access token*
- Select the below in Advanced Settings

Advanced settings

▼ Hide

Test Request URL

Provide a relative URL to test the connection, e.g. `/user/profile`. A GET request valid.

Object name field

Operation ID

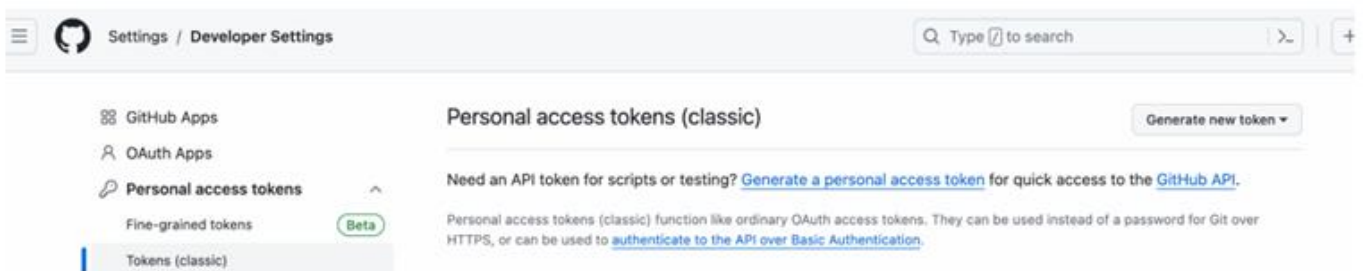
Select the field to use for the object names in the pick list.

Operation name field

Summary

Select the field to use for the operation names to execute in the pick list. Defai

- For those who do not have github account, they can create a Github account here – <https://shorturl.at/am1eL>
- Create a new repository with a readme file. For more details, refer to <https://shorturl.at/RgHY8>
- To create a personal access token, login to github, Navigate to Settings-> Developer Settings and click on generate a personal access token



- Select the access scopes as listed in the below screenshot and **create token**.

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events

☒ user	Update ALL user data
☒ read:user	Read ALL user profile data
☒ user:email	Access user email addresses (read-only)
☒ user:follow	Follow and unfollow users

- Once the connection has been created, create a recipe and test the below endpoint using the “Execute Operation” action

Operation *

Get the authenticated user

OAuth app tokens and personal access tokens (classic) need the `user` scope in order for the response to include private profile information.

16. Exercise steps for GraphQL Connector

- Install the Graph QL Connector from the Community library
 - <https://shorturl.at/3TdNL>
- Create a new Graph QL connection using the below settings. We will be using the same github instance for this exercise also.
 - GraphQL endpoint - <https://api.github.com/graphql>
 - Authentication method - Basic
 - Username - *Your github userID*
 - Password - *Personal Access token*
 - *Reuse the same access token created as part of the Open API exercise.*
- Create a recipe and use the “Run Operation” action and test the below query:

```

query {
  viewer {
    repositories(first: 10) {
      totalCount
      nodes {
        nameWithOwner
      }
      pageInfo {
        endCursor
        hasNextPage
      }
    }
  }
}

```

17. Exercise steps for SDK

Description: Acme Corp would like to validate the email of Contacts, Leads, etc before adding them to Salesforce. They would like to use WhoisXML API to do the email validation/verification.

Problem:

- No Connector for WhoisXML APIs.
- Universal HTTP actions have to be configured for every action for every use case which makes it not generic/reusable.
- Also the users of this connector do not have any technical understanding of invoking APIs.

Exercise steps for creating SDK Connector

- Use the Open API specification for creating your SDK connector.
- The Open API specification for WhoisXML Email Verification API is given below - <https://shorturl.at/pVjzt>
- Download the YAML file to your local system.
- Create a new SDK using the import OpenAPI specification option.
- *Name the connector starting with your*
CompanyName_GroupNumber_EMail_Validator

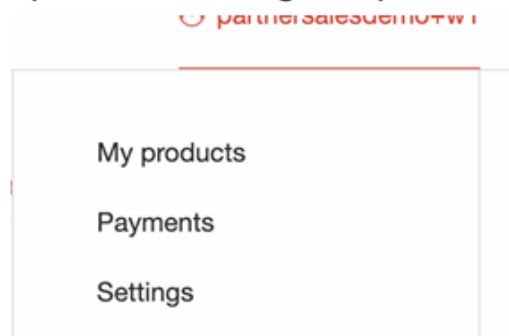
- Make sure to update line number 7 in the generated SDK code to params instead of Header as the WhoisXML service expects the APIkey in the query parameter.

```

1  {
2    title: "Email Verification API",
3    connection: {
4      authorization: {
5        type: "api_key",
6        apply: lambda do |connection|
7          params(:"apiKey" => connection['apiKey'])
8        end

```

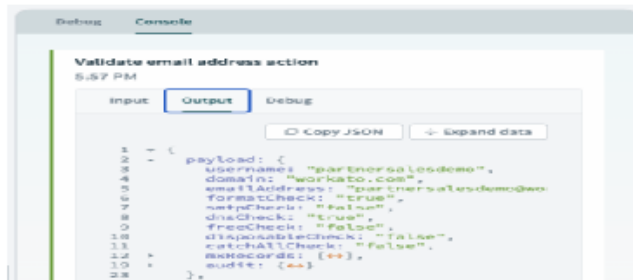
- Create a trial account with WhoisXML for obtaining API Key.
- Sign up for a trial account of WhoisXML API here – <https://whois.whoisxmlapi.com/signup>
- After signup, login into WHOisXML using the below Link <https://whois-history.whoisxmlapi.com/api/login>
- Click on your profile in the right top corner and select My products.



- Copy your API key from this page.



- Test the SDK connector on the SDK debug console for the "Validate Email Address" action and then release the SDK locally to the workspace upon successful testing.
- Use any valid email for testing and observe the output fields, the sample response is as shown below.



18. Exercise steps for AI By Workato

Description: Acme Corp aims to leverage Gen AI to aid users in addressing customer issues (sometimes expressed in languages other than English) Utilizing **AI By Workato Connector**, the message will be translated, analyzed, and summarized efficiently. Additionally, an automated response email will be generated to assure the customer that their ticket is being addressed. This initiative promises significant time savings for support users and an improved customer experience.

Exercise steps

- Create a new recipe with a scheduler trigger.
- The customer is from Spain and has sent an email in Spanish concerning a Product issue.
- The Spanish email can be downloaded from [here](https://shorturl.at/ti7VC) (<https://shorturl.at/ti7VC>)
- Load the Spanish email to a variable in Workato.
- Next use **AI By Workato Connector** and add the following steps to the recipe
 - Translate Text into English
 - Analyze text using the below **instruction** against the translated content from the step above.

Instruction: Evaluate the customer's tone based on the content and classify it into categories such as Positive, Neutral, Frustrated, Angry or Confused.

- Summarize the Email in 50 words using Summarize text action

- Finally, generate a draft email based on the summary. The email description is shared below.

Email Description: Compose an email in response to the user's complaint provided here [<Map generated Summary in earlier step here>](#). Include empathy in acknowledging the user's issue and assure them that the team is actively seeking a resolution. Address the sender according to the provided summary.

- Use the below flow in the recipe -

